

Problem

For an emergency situation, an engineer needs to make an RC high-pass filter. He has one 10-pF capacitor, one 30-pF capacitor, one 1.8-k Ω resistor, and one 3.3-k Ω resistor available. Find the greatest cutoff frequency possible using these elements.

Step-by-step solution

Step 1 of 1

$$\omega_c = \frac{1}{RC}$$

We make R and C as small as possible to achieve this we connect 1.8K Ω and 3.3K Ω in parallel so that

$$R = \frac{1.8 \times 3.3}{1.8 + 3.3} = 1.164 \text{ K}\Omega$$

We place the 10PF and 30PF capacitor in series so that

$$C = \frac{(10 \times 30)}{40} = 7.5 \text{ PF}$$

Hence

$$\begin{aligned} \omega_c &= \frac{1}{RC} = \frac{1}{1.164 \times 10^3 \times 7.5 \times 10^{-12}} \\ &= 114.55 \times 10^6 \text{ rad / s} \end{aligned}$$